

Lecture Notes for Math 372: Elementary Probability and Statistics

Last updated: August 25, 2026

Probability theory has a right and a left hand. On the right is the rigorous foundational work using the tools of measure theory. The left hand “thinks probabilistically,” reduces problems to gambling situations, coin-tossing, models of a physical particle. – Leo Breiman

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0 Tentative course outline

This course is a problem-oriented introduction to the basic concepts of probability and statistics, providing a foundation for applications and further study.

- **Weeks 1-3:** Introduction to probability theory
 - Experiments, events, sets, probabilities, random variables. Equally likely outcomes, counting techniques. Conditional probability. Independence. Bayes' theorem. (Sections: 2.1-2.5)
- **Weeks 4-5:** Discrete random variables (1.5 weeks)
 - Expected values, mean, variance, binomial distribution, Poisson distribution. Moment generating functions. (Sections: 3.1-3.6)
- **End of week 5: Midterm 1 (Friday, September 25)**
- **Weeks 6-7:** Continuous Random variables (1.5 weeks)
 - Uniform, exponential, gamma, and normal distributions. Intuitive treatment of the Poisson process and development of the relationship with gamma distributions. (Sections: 4.1-4.4)
- **Weeks 8-9:** Multivariate distributions
 - Calculation of probability, covariance, correlation, marginals, conditions. Distributions of sums of random variables and sampling distributions. Central limit theorem. (Sections: 1.1, 1.3, 1.4, 5.1-5.7)
- **End of week 9: Midterm 2 (Tentatively: Friday, October 23)**
- **Week 10:** Introduction to statistical estimation
 - Point and confidence interval estimation. Maximum likelihood, optimal, and unbiased estimators. Examples. (Sections 6.1, 6.2)
- **Weeks 11-12:** Large sample inference
 - Estimation (1.5 weeks): Types and comparison of estimators; sampling distributions for means/proportions, and their use in large sample estimation; sample size. (Sections 7.1, 7.2)
 - Hypothesis testing (1.5 weeks): Components of a test; significance and power; p-values; large-sample tests for means and proportions (Sections: 8.1-8.4)
- **Week 13:** Small sample inference
 - t-distribution, with applications to small sample estimation and testing; χ^2 and F distributions, with applications to inference about variances (Sections: 7.3, 7.4, 8.3)
- **Weeks 14-16:** Regression and χ^2 tests
 - Regression (1.5 weeks): Least squares, correlation coefficient, inference (Sections: 12.1-12.5)
 - χ^2 tests: multinomial distributions, contingency tables, goodness-of-fit (Sections: 14.1-14.3)
- **Last day of instruction: Wednesday, December 9**
- **Final Exam: December 18**

1 2026-08-24 | Week 01 | Lecture 01

- give syllabus
- did the pirates worksheet

2 2026-08-26 | Week 01 | Lecture 02

The nexus question of this lecture: What is a set and what are the basic set operations?

Reading assignment: chapter 2.1

2.1 Set theory

The mathematical formalization of modern probability theory rests on the foundation of set theory.

2.1.1 Definitions and set-builder notation

- A **set** is an unordered collection of distinct objects. The objects of a set are called **elements**. We write the elements of a set inside curly braces, $\{\dots\}$.

Examples: $\{1, 4, 2, 3\}$, $\{\text{cat}, \text{fish}, 5\}$, $\mathbf{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$, $\mathbf{R} = \{\text{the real numbers}\}$

- Two sets A and B are **equal**, denoted $A = B$, if they share precisely the same elements. The order that the elements are written is irrelevant.

$$\{1, 2, 3\} = \{3, 1, 2\}.$$

Repeated elements are ignored:

$$\{1, 2, 3\} = \{1, 1, 2, 3\}.$$

- A set is **finite** if it contains a finite number of distinct elements. This number is called the **size** of the set, and is denoted $|A|$.

Example: $|\{\text{cat}, 5\}| = 2$.

- The **empty set** is the set which contains no elements. This is denoted as

$$\emptyset \quad \text{or} \quad \{\}$$

- **Set-builder notation** describes a set by specifying the properties that its elements must satisfy.

Examples:

The set of integers between 1 and 36:

$$\{n \in \mathbb{Z} : 1 \leq n \leq 36\}$$

The set of ordered pairs (x, y) such that x and y are both positive real numbers:

$$\{(x, y) \in \mathbb{R}^2 : x, y > 0\}$$

The set of even integers:

$$\{n \in \mathbb{Z} : n \text{ is even}\}$$

The set of ordered pairs x, y such that both x and y are integers between 1 and 4:

$$\{(x, y) : x, y \in \{1, 2, 3, 4\}\}$$

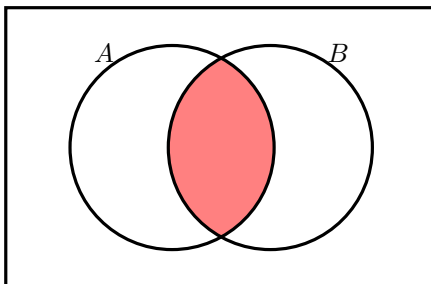
In set builder notation, the colon ‘:’ is read as “such that” and the symbol ‘ $\in$ ’ (which looks like an E) is read as “is an element of”

2.1.2 Set operations

Let A, B be sets.

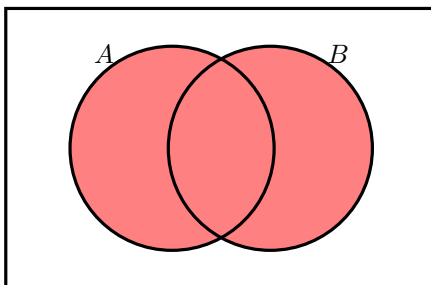
- The **intersection** of A and B , denoted $A \cap B$, is defined by

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$



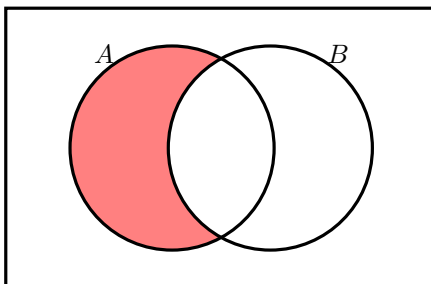
- The **union** of A and B , denoted $A \cup B$, is defined by

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$



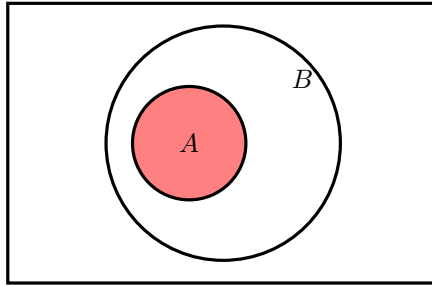
- The **set difference** of A and B , denoted $A \setminus B$, is

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$



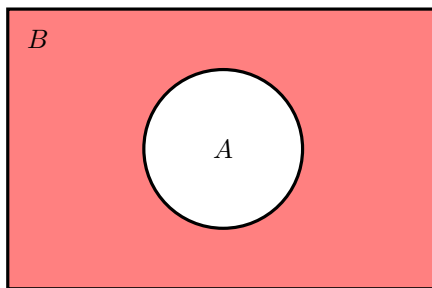
- We say that A is a **subset** of B , and write $A \subseteq B$, if

$$x \in A \implies x \in B.$$



- If $A \subseteq B$, the *complement of A in B*, denoted A^c , is

$$A^c = B \setminus A = \{x : x \in B \text{ and } x \notin A\}.$$



- We say that A and B are *disjoint* if $A \cap B = \emptyset$.

